Quantum Computing Foundations

Lab Handout

Session Overview

Duration: 1h15m | **Format:** Run notebook cells step-by-step during lecture.

Session Objectives

- Describe the difference between classical bits and qubits, and write a single-qubit state as $\alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$.
- Explain how two qubits span four basis states and interpret a general 2-qubit state $|\Psi\rangle$ in Dirac and vector form.
- Use Qiskit to construct simple circuits that create superposition (via the Hadamard gate) and visualize the resulting measurement histograms.
- Build and simulate a Bell-state circuit (H + CNOT), and interpret the resulting correlated outcomes as evidence of entanglement.
- Identify the roles of common gates (H, X, CNOT, measurement) and relate their action to changes in state and measurement probabilities.
- Discuss qualitatively how interference and decoherence affect quantum computations and histograms in simulator-based experiments.

Timed Agenda:

- 0-10min: Classical vs Quantum Basics
- 10-30min: Qubits & Superposition
- 30-50min: Entanglement & Gates
- 50-70min: Measurement & Interference
- 70-75min: Challenges & Wrap-up

1. Setup (Run First - 2min)

```
!pip install qiskit qiskit-aer matplotlib  # uncomment if needed
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator
from qiskit.visualization import plot_histogram, plot_bloch_multivector
import matplotlib.pyplot as plt
import numpy as np
simulator = AerSimulator()
print('Qiskit ready!')
```

2. Classical Bits vs Qubits (10min)

Classical bit: {0,1}.

Qubit: $\alpha|0\rangle + \beta|1\rangle$ where $|\alpha|^2 + |\beta|^2 = 1$.

2 qubits explore all 4 states in parallel:

$$|\Psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

3. Superposition (20min)

Basic States & Matrices

State	Vector Form
$\|0\rangle$	$\begin{pmatrix}1\\0\end{pmatrix}$
$\|1\rangle$	$\begin{pmatrix}0\\1\end{pmatrix}$
$\|+\rangle = H\|0\rangle$	$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$
$\|-\rangle = H\|1\rangle$	$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$
Hadamard Gate (H)	$\frac{1}{\sqrt{2}}\begin{pmatrix}1&1\\1&-1\end{pmatrix}$
X Gate (NOT)	$\begin{pmatrix}0&1\\1&0\end{pmatrix}$

The Hadamard gate H transforms the computational basis state $|0\rangle$ into the superposition state $\frac{|0\rangle+|1\rangle}{\sqrt{2}}$.

State Vectors Computational basis states:

$$|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \quad |1\rangle = \begin{pmatrix}0\\1\end{pmatrix}.$$

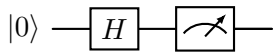
Hadamard Matrix

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix}1&1\\1&-1\end{pmatrix}.$$

Matrix Calculation

$$H|0\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\cdot 1 + 1\cdot 0\\ 1\cdot 1 + (-1)\cdot 0\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle\,.$$

Quantum Circuit



Hadamard: $|0\rangle \to \frac{|0\rangle+|1\rangle}{\sqrt{2}}$

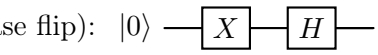
Qiskit

```
# Quantum superposition
qc = QuantumCircuit(1, 1)
qc.h(0)  # Hadamard = equal superposition
qc.measure(0, 0)
```

```
# Run 1000 shots
result = simulator.run(transpile(qc, simulator), shots=1000).result()
plot_histogram(result.get_counts(), title='Superposition: ~50/50')
```

```
qc_bloch = QuantumCircuit(1)
qc_bloch.h(0)
plot_bloch_multivector(qc_bloch)  # Equator of Bloch sphere
```

Variation:

X then H (phase flip): 

4. Entanglement (20min)

Tensor Product & Bell States

Tensor product: The tensor product combines two quantum states into a joint state on the tensor product Hilbert space. For single-qubit states $|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}$ and $|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}$, it stacks the second vector for each entry of the first.

Computation The tensor product $|0\rangle \otimes |0\rangle$ is computed as follows:

$$|0\rangle \otimes |0\rangle = \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\cdot\begin{pmatrix}1\\0\end{pmatrix}\\ 0\cdot\begin{pmatrix}1\\0\end{pmatrix}\end{pmatrix} = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}.$$

This 4D vector lives in $\mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$ and corresponds to the binary index 00.

Comparison to Example Consider the example $|+\rangle \otimes |0\rangle$, where $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$:

$$|+\rangle\,|0\rangle = |+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}1\\0\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\0\end{pmatrix}.$$

Here, $|0\rangle \otimes |0\rangle$ has no superposition and full amplitude on the top entry.

State	**Vector Form (4D)**
$\|00\rangle$	$\begin{pmatrix}1\\0\\0\\0\end{pmatrix}$
$\|11\rangle$	$\begin{pmatrix}0\\0\\0\\1\end{pmatrix}$
$\|\Phi^+\rangle = \frac{\|00\rangle+\|11\rangle}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\1\end{pmatrix}$
$\|\Phi^-\rangle = \frac{\|00\rangle-\|11\rangle}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\-1\end{pmatrix}$
CNOT (q0→q1)	$\begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix}$

Step-by-Step Creation

$$|00\rangle \xrightarrow{H\otimes I} |+\rangle\,|0\rangle = \frac{|00\rangle + |10\rangle}{\sqrt{2}} \xrightarrow{\text{CNOT}} |\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

The sequence describes the standard Bell state preparation circuit starting from $|00\rangle$, applying Hadamard on the first qubit ($H \otimes I$), followed by CNOT (control on first, target on second) to create the entangled Bell state $|\Phi^+\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$.

Initial State

The initial state in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ is:

$$|00\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}.$$

$H \otimes I$ Matrix

Hadamard gate: $H = \frac{1}{\sqrt{2}}\begin{pmatrix}1&1\\1&-1\end{pmatrix}$, Identity: $I = \begin{pmatrix}1&0\\0&1\end{pmatrix}$. The tensor product is:

$$H \otimes I = \frac{1}{\sqrt{2}}\begin{pmatrix}1&0&1&0\\0&1&0&1\\1&0&-1&0\\0&1&0&-1\end{pmatrix}.$$

Applying to $|00\rangle$ yields $|+0\rangle = \frac{|00\rangle+|10\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\1\\0\end{pmatrix}$.

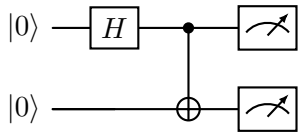
CNOT Matrix

Standard CNOT (control qubit 1, target qubit 2):

$$\text{CNOT} = \begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix}.$$

Applying to $|+0\rangle$ produces $|\Phi^+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\1\end{pmatrix}$.

Quantum Circuit



Qiskit Verification

```
# Perfect Bell pair
bell = QuantumCircuit(2, 2)
bell.h(0)
bell.cx(0, 1)
bell.measure([0,1], [0,1])
```

```
result_bell = simulator.run(transpile(bell, simulator), shots=1000).result()
plot_histogram(result_bell.get_counts())  # Only 00, 11 (~50% each)
```

5. Quantum Gates Summary (15min)

Gate	Symbol	Effect
H	Hadamard	Superposition
X	NOT	$\|0\rangle \to \|1\rangle$
CNOT	Controlled-X	Entangles qubits
Measure	M	Collapse to classical

6. Interference & Challenges (10min)

Interference: Amplitudes add (+)/cancel (-).

Challenges: Decoherence destroys superposition; needs error correction.

Homework:

Add H to qubit 1 before CNOT. Predict histogram?

```
# Student modification
your_circuit = QuantumCircuit(2,2)
# your_circuit.h(1)  # Try this!
your_circuit.h(0)
your_circuit.cx(0, 1)
your_circuit.measure_all()
```

Key Takeaways

- **Superposition** = $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$ (parallelism across all states)
- **Entanglement** = $|\Phi^+\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$ (perfect correlations (00 or 11 only))
- **Measurement** = probabilistic collapse ($\sim$50/50 histograms)
- **H + CNOT** = universal 2-qubit recipe